

Print log

2026/03/24

03/24 - RUN 1

- vase shape:
 - D-50mm,
 - H-350mm
 - OUTWARD - 50mm
 - vertical profile angle: 45 degrees
- Set vel: 100 mm/s,
- extr rpm: 400,
- Override: 30%, gradually, from 10 to 50 during trying
- Material
 - LX 175 PLA
 - 1% TiO2, white masterbatch
- temp: 190 degrees,
- nozzle size: 3mm,
- set layer height: 1mm
- idea section:
 - width: 3mm,
 - layer height: 1mm
- material use:
 - start: 20:23, 1st 200g
 - 2nd: 20:48, 2nd 200g
- src lines: 77k

issue:

- the print bed may be extruded by the heat of printout
- the melt printout is not fully cool down, so the printed part is tackled by the nozzle movement
- the layer height might be too low for 1.0mm

advice:

参数	当前值	建议值	原因
层高	1.0mm	1.5~2.0mm	更好的宽高比，结构更稳定
挤出机RPM	400	按体积比例调整	层高增大后需匹配流量
速度/Override	30% (30mm/s)	20~25% 或加层间等待	给每层更多冷却时间
APO_DIST	50mm	10~15mm	更好的花瓣形状保真度
层结构	离散	螺旋连续Z	消除Z缝，更均匀
冷却	无	加风扇/层间暂停	关键改进
平台	纸板/MDF	金属板+隔热垫	防翘曲

03/24 - RUN 2

- vase shape:
 - D-50mm,
 - H-350mm
 - outward- 35mm
 - vertical profile: 30.9 degrees
- Set vel: 50 mm/s,
- extr rpm: 400,
- Override: 24%
- Material
 - LX 175 PLA
 - 1% TiO2, white masterbatch
- temp: 185 degrees,
- nozzle size: 3mm,
- set layer height: 1.3mm (1st layer: 1.5)
- section:
 - set width: 3mm
 - act width: 6.92mm
- material use:
 - start: NAN, but almost 200g there
 - 2nd: 21:56, 200g, the last amount is approx 1/5, with air vibrator on
 - 22:11, left around 200g
 - ☐ 15min. 200g?

- 3rd: 22:14, 200g
- 4th: 22:16, 200g, 600g now inside the hopper
 - ☐ Hopper max volume: 800g?
- 5th: 11:44, 200g, last approx 600g almost run out
 - cal: 600g/88min, 6.81g/min
 - cal: if count as 200/15min, 13.33g/min
 - **need a more careful calib: refer to deposition efficiency inside barrel**
- 6th: 12:15, 200g, approx 200g/20min
- Finish: 12:25, only consumes 1/10 of 6th round of 200g
- total: 1.0+ kg
- src lines: 45k

issue:

- ✓ 2nd ptp causes collision
- ✓ already fixed with header replace script
- ✓ the tackling issue still there, but much more slighter
- ✓ with fan on the side, not obvious touching move

